

GTU Automated Circular & Notification Alert System

Real-Time University Notice Broadcast with Zero-Latency & Direct 1-Click Official PDF Links

PROJECT SCOPE:	First Year Engineering Project (Computer / IT Dept)
TEAM LEADER:	Muhammad Sambhyana (Lead Developer & System Architect)
TEAM MEMBERS:	Saad Mansuri, Zakwan Nawab, Sunny Gondaliya, Prit Patel, Ved Patel, Shivans Tiwari, Anmol Varma, Khushi Gohil
CORE TECH STACK:	Python 3, BeautifulSoup4, SQLite3, Telegram API, Web PWA
DATA PRIVACY POLICY:	100% Public Notice Board Only • Zero Student Data Access

■ SECURITY & ETHICAL GUARANTEE:

This project strictly monitors **open public notices** published at gtu.ac.in. It does **NOT** require, extract, store, or transmit any student credentials, enrolment numbers, passwords, or private information.

1. Problem Statement & Solution Vision

Why students miss critical deadlines and how automated notification bridges the gap.

■ Current Challenges & Inefficiencies

- **Late Fee Penalties:** Exam registration & penalty deadlines are frequently missed due to lack of timely alerts.
- **Manual Friction:** Visiting gtu.ac.in daily on mobile devices is tedious and easily forgotten.
- **Information Overload:** 10-20 mixed notices are released daily; filtering branch-relevant news is difficult.
- **Faculty Burden:** Class coordinators manually download & forward circular PDFs in multiple WhatsApp groups.

■ The Automated Solution

- **Instant Mobile Delivery:** Scans GTU public board every 15 min & broadcasts alerts directly to Telegram.
- **Direct 1-Click PDF Link:** Students can view official circulars instantly with one tap inside the message.
- **Smart Urgency Tagging:** Distinct visual badges for Fee Deadlines (■), Timetables (■), and Results (■).
- **24/7 Cloud Autonomy:** Continuous monitoring via GitHub Actions with 0 server maintenance cost.

Zero Delay

Push updates within minutes of public posting.

Zero Spam

SQLite deduplication guarantees 1 post per notice.

Zero Cost

Free cloud execution & open-source Python stack.

■■ Presentation Script (Talking Point for Faculty):

'Respected Sir/Ma'am, students often face heavy exam penalties due to missing manual checks on GTU's site. Our system bridges this gap by automatically detecting new notices the moment GTU uploads them and sending rich alerts to student groups.'

2. Strict Data Privacy & Ethical Compliance

Absolute assurance regarding zero private student data access and safe university server usage.

■■ 100% ZERO STUDENT DATA ACCESS POLICY

This application acts strictly as an **Automated Public Reader**. It does **NOT** require, extract, or store any student enrolment number, password, name, phone number, or academic record from GTU or any student.

Data Domain	Access Status	Technical Reason & Compliance Details
Student Personal Data	ZERO ACCESS	No enrolment numbers, names, emails, mobile numbers, or passwords are ever requested or saved.
Login & Auth Portals	NEVER ACCESSED	No access to student.gtu.ac.in or ERP logins; no CAPTCHA bypass attempted.
Information Source	PUBLIC ONLY	Only reads public circulars table at gtu.ac.in/Circular.aspx visible openly to everyone.
Data Extracted	PUBLIC METADATA	Only extracts 3 public fields: Circular Title, Publication Date, and Official Public PDF Download URL.
GTU Server Load	RATE-LIMITED	Safe 15-minute polling interval with standard User-Agent headers creates less load than 1 human user.

■■ Presentation Script (Talking Point for Privacy Defense):

'Sir/Ma'am, we specifically ensured that our bot strictly acts as a public broadcaster. It operates without any logins or student personal records. It only retrieves what GTU has already made publicly accessible for all students on its website.'

3. System Architecture & Smart Features

High-level execution workflow and intelligent category tagging engine.

Step 1: Public Web Scraper Fetches gtu.ac.in/Circular.aspx using BeautifulSoup parser.	Step 2: Category Classifier Identifies Fee deadlines, Exam notices, Results, & Timetables.
Step 3: SQLite Deduplication Checks circulars.db to discard previously notified records.	Step 4: Push Broadcast Sends rich HTML message with 1-click PDF link to Telegram group.

Intelligent Urgency Categorization Badges

■ Fee & Penalty Alerts Highlights regular/remedial exam fees & penalty dates.	■ Exams & Timetables Mid-sem & End-sem timetables, hall tickets, and centres.
■ Result Announcements Instant alerts for declared semester & recheck results.	■ Academic & Admission Syllabus changes, academic calendar, and enrollment.

Core Language	Python 3.10+ (Standard Web Scraping & Async Libraries)
Web Parsing	BeautifulSoup4 + Requests with custom HTTP Header handlers
Database Engine	SQLite3 (Zero-latency local deduplication storage)
Cloud Infrastructure	GitHub Actions Cloud Cron (24/7 Free Scheduled Autonomous Runner)

4. Real-World Impact & Viva Defense FAQ

Quantifiable benefits and prepared defenses for faculty viva questions.

Student Benefits

- Zero late fee penalties.
- Instant PDF access without slow web navigation.
- Mobile-friendly push notifications.

College & Faculty Benefits

- Eliminates manual PDF forwarding by CRs/Staff.
- Higher on-time exam form submission rates.
- Demonstrates campus workflow automation.

Frequently Asked Questions (Viva Defenses)

Q1: Does this scraper risk crashing GTU's web servers?

Answer: No. The bot makes only 1 lightweight HTTP GET request every 15 minutes, consuming fewer server resources than a single student refreshing the page.

Q2: Is there any risk of student confidential data breach?

Answer: Zero risk. The bot never requests or stores enrolment numbers, passwords, or personal credentials. It only extracts public circular URLs.

Q3: What if the hosting machine loses internet connection?

Answer: The project includes an automated GitHub Actions cloud workflow that runs 24/7 on GitHub's cloud servers independently.

■ **Future Scope & Roadmap:** (1) **WhatsApp Cloud API** multi-channel broadcast. (2) **Branch & Semester filtering** (Computer/IT/Mech specific channels). (3) **AI Summary Engine (TL;DR)** for 1-sentence key takeaway summaries of lengthy PDFs.